

More or Less? Effects of Visual Information Modulation on Context Perception

Jifan Yang
Xi'an Jiaotong-Liverpool University
Suzhou, China
Jifan.Yang23@student.xjtlu.edu.cn

Fuqi Xie
Xi'an Jiaotong-Liverpool University
Suzhou, China
fuqi.xie20@student.xjtlu.edu.cn

Ziteng Li
Beijing Institute of Technology
Beijing, China
li0020210@163.com

Bo Fan
Xi'an Jiaotong-Liverpool University
Suzhou, China
Bo.Fan24@student.xjtlu.edu.cn

Yihan Liu
Xi'an Jiaotong-Liverpool University
Suzhou, China
yihan.liu17@student.xjtlu.edu.cn

Zhaolin Lu
Beijing Institute of Technology
Beijing, China
zhaolin.lu@bit.edu.cn

Yu Liu
Department of Computing
Xi'an Jiaotong-Liverpool University
Suzhou, China
Yu.Liu02@xjtlu.edu.cn

Min Chen
Xi'an Jiaotong-Liverpool University
Suzhou, China
min.chen@xjtlu.edu.cn

Martijn ten Bhömer
Xi'an Jiaotong-Liverpool University
Suzhou, Netherlands
Martijn.Tenbhommer@xjtlu.edu.cn

Eng Gee Lim
Xi'an Jiaotong-Liverpool University
Suzhou, China
enggee.lim@xjtlu.edu.cn

Lingyun Yu*
Department of Computing
Xi'an Jiaotong-Liverpool University
Suzhou, China
Lingyun.Yu@xjtlu.edu.cn

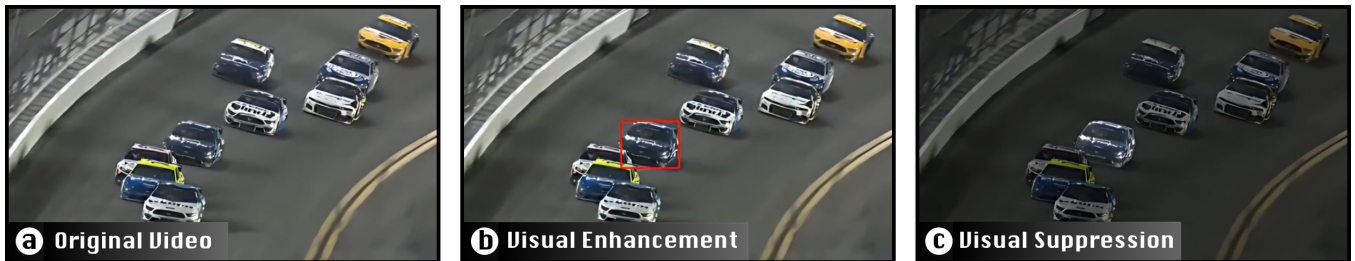


Fig. 1: Three video types: (a) Original Video, the unaltered, original footage. (b) **Visual Enhancement**, bounding boxes highlight key objects. (c) **Visual Suppression**, key object remains clear, surroundings dimmed to reduce distractions.

Abstract

This study examines the effects of two visual guidance techniques, **Visual Enhancement** and **Visual Suppression**, on user perception of contextual information in video content. **Visual Enhancement** introduces explicit visual cues to highlight target content, whereas **Visual Suppression** attenuates non-target elements, for example, by reducing their brightness. Both approaches aim to isolate specific objects from the background, directing attention to

critical information within complex, dynamic scenes. Despite their growing usage, the relative effectiveness of these approaches in guiding attention and their impact on peripheral context awareness remain underexplored. To address this gap, we conducted a controlled user study with 27 participants. The results indicate that **Visual Enhancement**, through the addition of salient cues, more effectively directs user attention to target information than **Visual Suppression**. Our findings advance understanding of visual attention in dynamic environments and offer implications for designing visual guidance strategies.

*Corresponding author



This work is licensed under a Creative Commons Attribution 4.0 International License.
VINCI 2025, Linz, Austria

© 2025 Copyright held by the owner/author(s).
ACM ISBN 979-8-4007-1845-8/25/12
<https://doi.org/10.1145/3769534.3769568>

CCS Concepts

• Human-centered computing → Empirical studies in visualization.

Keywords

visual guidance, visual representation design

ACM Reference Format:

Jifan Yang, Fuqi Xie, Ziteng Li, Bo Fan, Yihan Liu, Zhaolin Lu, Yu Liu, Min Chen, Martijn ten Bhömer, Eng Gee Lim, and Lingyun Yu. 2025. More or Less? Effects of Visual Information Modulation on Context Perception. In *Proceedings of the 18th International Symposium on Visual Information Communication and Interaction (VINCI 2025)*, December 01–03, 2025, Linz, Austria. ACM, New York, NY, USA, 5 pages. <https://doi.org/10.1145/3769534.3769568>

1 Introduction

In an era increasingly driven by visual media, individuals are constantly exposed to vast amounts of visual data, yet their capacity to process such data remains limited. To direct attention to important details, prior research has examined two categories of visual guidance techniques. The first approach is to **enhance imagery with additional elements**, such as bounding boxes [21] and arrows [12]. This method makes key information prominent but inevitably increases visual complexity [36]. The second approach is to **suppress less relevant information**, typically by reducing luminance [6], saturation [5], or clarity [3]. This technique highlights the target without introducing external cues but risks obscuring contextual details. Thus, both strategies require careful balancing to avoid either imposing cognitive overload or underinforming the user.

Beyond focusing on a target, individuals also strive to stay aware of the broader context. For instance, racing spectators may concentrate on their favored competitor while seeking to be aware of key events such as collisions and overtakes in the background. Similarly, film audiences may focus on lead actors while relying on the supporting cast and set design to follow the narrative. This dual demand raises two important questions: First, which approach—**enhancing primary information** or **suppressing secondary information**—assists users in directing their attention to their observation target with lower cognitive load? Second, when either technique is employed, does the emphasis on specific objects impair users’ awareness and understanding of the broader context?

To explore these questions, we conducted a user study on how visual guidance techniques shape users’ perception of video content. We developed a processing tool that applies two representative techniques: **highlighting key objects with bounding boxes (enhance)** and **dimming less essential information (suppress)**. Participants viewed both original and processed videos, and we compared their perceived workload and comprehension of the target and context. Based on the results, we derived design considerations and a broader discussion of visual guidance methods. Overall, our work makes the following contributions:

- A video processing tool for applying **Visual Enhancement** and **Visual Suppression** techniques.
- A user study evaluating their impact on video perception.
- Practical guidelines for visual information modulation.

2 Related Work

In this section, we review prior research on visual attention mechanisms and cognitive limitations that impact context perception.

2.1 Visual Attention Mechanisms

Human visual processing is inherently limited, making it difficult to perceive all elements with equal precision. Instead, individuals depend on selective attention [17], which is guided by two mechanisms [14]: bottom-up attention attracted by visual input, and top-down attention directed by goals and prior knowledge.

This study focuses on the bottom-up mechanism, which is driven by visual salience [15]—the prominence of elements within a scene. Previous studies have extensively examined visual features that facilitate bottom-up attention, finding that strong color or luminance contrasts [37], bright regions [11], clear versus blurred areas [8], and enclosed regions [28] can involuntarily draw quick and frequent fixations. These findings have informed the design of **Visual Enhancement** and **Visual Suppression** techniques, which seek to direct attention by enhancing the relative salience of key elements in complex visual environments. Notably, individuals can override bottom-up attention through top-down control, ignoring stimuli they deem irrelevant [9]. While effective for visual search, this process introduces variability across users and contexts. In our work, we reduce this variability to better assess how bottom-up salience influences attention by standardizing the attention target.

2.2 Inattentional Blindness

Selective visual attention, while aiding cognitive load management, reduces the processing of contextual information [40], especially in complex environments demanding exceed attentional capacity [23]. A frequently reported consequence is inattentional blindness [25], where individuals concentrating on a specific target fail to notice other salient stimuli in their visual field. A well-known example comes from Simons and Chabris’s gorilla experiment [33], where participants tracking basketball passes often failed to notice a person in a gorilla suit walking through the scene. Koivisto et al. [19] further noted that inattentional blindness occurs regardless of the observer’s fixation point—whether gazing at the center or periphery of the scene, contextual stimuli remain difficult to detect. These findings raise an important question: while visual guidance techniques aim to direct attention to specific targets, might they inadvertently increase neglect of important contextual elements? To address this, our study examines the impact of two representative techniques on user perception and comprehension.

3 Overview

In this section, we introduce existing visual guidance approaches and the rationale for selecting the techniques used in our study.

Visual Enhancement enhances target salience without altering context through either external cues or internal modifications. External cues progress in precision: glyphs indicate a direction [12, 20] (Fig.2 (a)); bounding boxes highlight a rough area [7] (Fig.2 (b)); and glows or contours precisely delineate boundaries [1, 38] (Fig.2 (c)). Internal modifications instead alter the target itself, such as adding overlays (e. g., color [32], patterns [1]) or changing visual properties [26, 30] (Fig.2 (d)). Temporal dynamics (e. g., flicker) can be combined with either form to further enhance salience [31, 35].

Visual Suppression is context-focused, drawing attention by reducing the salience of surroundings while leaving the target unchanged. Suppression can also be broad or follow the target’s exact boundaries, typically via lowered saturation, luminance, transparency (Fig.2e), or applying blur (Fig.2f) to contextual regions [16].

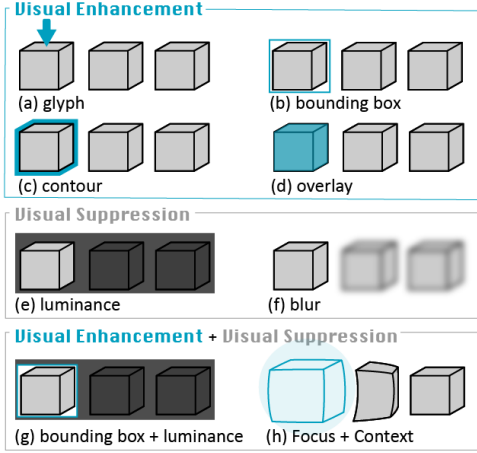


Fig. 2: Visual guidance techniques. **Visual Enhancement:** (a) glyph indicator, (b) bounding box, (c) precise outline, and (d) color mask. **Visual Suppression:** (e) luminance reduction and (f) blur. **Visual Enhancement + Visual Suppression:** (g) bounding box + luminance and (h) Focus + Context.

Visual Enhancement and **Visual Suppression** can be combined to further emphasize the target. A common example is the spotlight effect [21], which outlines the target while dimming the background (Fig.2g). Another is Focus+Context, where the target region is magnified and the surroundings are compressed (Fig.2h).

Our study explores two representative techniques—**bounding box** (Fig.2 (b)) around target object and **luminance reduction** (Fig.2 (e)) of non-target information—both reflects real-world practice. Bounding boxes are widely used in object detection [21], while luminance adjustment is common in instructional videos [6]. Notably, they leverage different visual mechanisms—color and boundary cues versus brightness contrast. Our goal is not to compare methods with strictly matched salience, but to evaluate the effectiveness of commonly used approaches.

4 The Video Processing Tool

We developed a video processing tool for the selected techniques. The workflow, shown in Fig. 3, includes the following steps:

Frame Extraction. The video is processed using OpenCV, which breaks it down into individual frames. The first frame is designated as the reference frame for later object detection. Simultaneously, the audio track is extracted for later synchronization.

Target Selection and Object Segmentation. The target object is manually selected from the reference frame. The SAM model [18] is employed for visual segmentation, generating a binary mask that isolates the target from the background. The mask is then passed to the DeAOT framework [4, 41], which tracks the target across subsequent frames and produces a sequence of corresponding masks.

Visual guidance and Integration. Based on the binary masks, the selected visual guidance technique—a **bounding box** or luminance adjustment—is applied to each frame. The frames are re-assembled into a video, with the original audio track synchronized.

Optimization. The processed video can be further refined, such as re-selecting the target object, selecting additional objects for multi-object tracking, or designating new reference frames. These adjustments trigger re-segmentation and re-tracking of targets.

5 User Study

We conducted a user study using videos edited with the processing tool to evaluate user perceptions under visual guidance, approved by the XJTLU Ethics Review Panel (Protocol № 20250112001828).

Materials. We selected racing videos as the experimental material, as they typically feature **complex and dynamic visuals**, **high attentional demand**, and **minimal compositional bias**. Moreover, the simple rules of racing can promote shared understanding among participants and provide clear criteria for evaluation. We extracted a 36-second NASCAR racing video [27] featuring 15 cars and a crash event (Fig. 4). Three video versions were created: **Original Video**, **Visual Enhancement** video, and **Visual Suppression** video (Fig. 1). Car No.17 (Fig. 4c) was chosen as the target in all videos due to its consistent visibility and relative isolation. Vehicles involved in key events, such as the crashed car and the race winner, served as cues to assess participants' contextual awareness.

Pilot study. A pilot study with 3 participants was conducted to optimize background luminance in the **Visual Suppression** condition. After iterative adjustments, the luminance was set to 61%, balancing target clarity and overall scene visibility.

Participants. We recruited 27 participants (14 males and 13 females) from two universities, providing 80% power ($\alpha = .05$) to detect within-subject effects of Cohen's $f \approx 0.25$. All participants had normal color vision and normal or corrected-to-normal visual acuity. Five participants had no experience with similar videos, 10 watched racing sports annually, 6 monthly, 4 weekly, and 2 daily.

Task and Procedures. Participants each completed three sessions, viewing one video version (**Original**, **Visual Enhancement**, or **Visual Suppression**) per session on a 13-inch MacBook Air, with presentation order counterbalanced. Before each session, they were shown a photograph of the target vehicle and asked to track it. After each session, they completed questionnaires on perception and viewing experience. To reduce learning effects, consecutive sessions were separated by a 5-day washout period in accordance with memory retention research [22, 34]. Participants further confirmed no recollection of their previous responses.

Measures. Participants' perception of video content was assessed using multiple-choice questions covering general context, the target, and key competitors (see Appendix for item details). Perceived workload of tracking the target was measured with a 7-point adaptation of the NASA-TLX [10], which has been validated in prior studies (e. g., [24, 29]). Responses were analyzed using bootstrapped 95% CIs and RM-ANOVA with Bonferroni corrections.

6 Results

As shown in Fig.5, participants reported significantly lower overall workload when tracking the target in the **Visual Enhancement** condition compared to **Original Video** ($p = .019$), with reductions in mental, physical, temporal, and effort loads, lower frustration, and slightly higher perceived performance. The **Visual Suppression** condition exhibited lower workloads than **Original Video**, with similar performance scores but slightly higher frustration; however, these differences were not significant ($p = .445$).

Mean accuracy across video-related questions is shown in Fig.6. Participants showed high accuracy for general context across all conditions. For observation target, **Visual Enhancement** videos

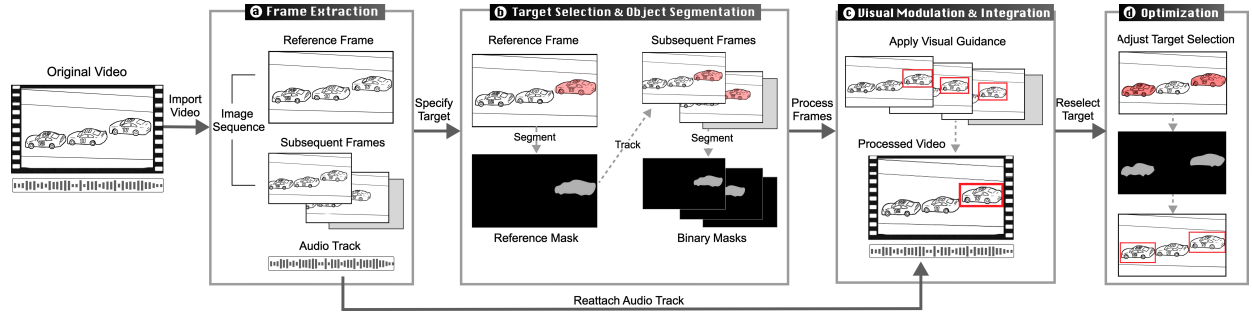


Fig. 3: The tool processes videos in four steps: (a) extract the first frame; (b) segment and track the target object; (c) apply visual effects and generate output; (d) adjust selection and update results.



Fig. 4: The video used in the user study consists of: (a) 15 race cars speeding on the track. (b) A car crashes and leaves the track. (c) Car No. 17 is designated as the primary focus target.

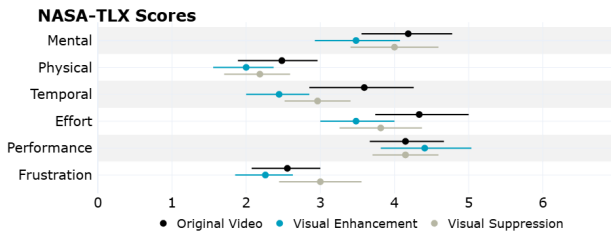


Fig. 5: Mean scores and 95% CIs of workload and performance.

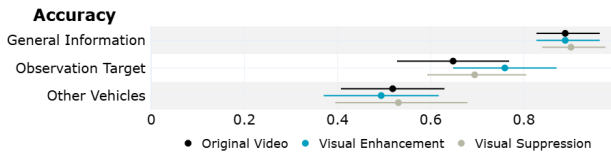


Fig. 6: The mean accuracy and 95% CIs in question responses.

produced higher accuracy than **Original Video** and **Visual Suppression**, but the difference was not significant ($p = .623$). Accuracy for other vehicles remained low across all conditions, indicating limited attention to contextual details. Overall, **Visual Enhancement** enhanced accuracy for the target but slightly reduced accuracy for other vehicles, whereas **Visual Suppression** and **Original Video** yielded similar performance across question types.

7 Discussion

Design Recommendations. Focusing on a target reduces contextual awareness, and explicit cues like **[bounding boxes]** may amplify this effect. Visual guidance should thus be applied **only when necessary and with moderate emphasis**, avoiding overly intrusive cues or excessive suppression. Second, **distinct target boundaries** could facilitate tracking, as observed in the **Visual Enhancement** condition, while subtle demarcation in **Visual Suppression** may require more effort. Finally, visual guidance should **account for individual differences**, adapting enhancement or suppression to user behavior and preference.

Other Visual Guidance Techniques. While this study focuses on two representative techniques, other techniques in Fig. 2 face similar trade-offs in balancing target and context awareness. Overlays, glows, and Focus+Context methods can obscure details or distort visuals, while glyphs may increase cognitive effort in interpretation. Therefore, the design recommendations outlined above can be extended to guide the development of these approaches.

Visual Guidance for Higher-Level Perception. **Visual Enhancement** methods can further incorporate visuals such as structural overlays [32], color gradients [2], glyphs [13], or textual annotations [39] to help users understand spatial, temporal, or functional relationships. **Visual Suppression** approaches, while often more subtle, can facilitate comparison, pattern recognition, and hypothesis formation. Together, these techniques provide complementary strategies for augmenting both attention and comprehension.

8 Limitation and Conclusion

Addressing the first research question—whether **Visual Enhancement** or **Visual Suppression** better supports target focus—we found that participants in the **Visual Enhancement** condition demonstrated higher accuracy and lower cognitive effort. For the second question—whether users can grasp the surrounding context—while participants retained a general sense of the background, their awareness of finer details was limited, resulting in incomplete understanding. These findings highlight the need to emphasize key elements without obscuring critical context.

We acknowledge several limitations that merit further investigation. This study focused on two visual guidance techniques within the context of racing videos, which may limit the generalizability of findings. In addition, participants' prior familiarity with video content was not controlled, and we did not conduct fine-grained comparisons of visual attributes, such as bounding box color, opacity, or suppression level. Furthermore, our assessment relied on self-report and multiple-choice measures. Future research could integrate quantitative methods, such as eye tracking, with qualitative feedback to provide a more comprehensive evaluation.

Acknowledgments

Lingyun Yu was partially supported by the National Natural Science Foundation of China (Grant № 62272396). Zhaolin Lu was partially supported by the National Natural Science Foundation of China (Grant № 52275234) and Humanities and Social Science Fund of the Ministry of Education of China (Grant № 22YJA760055).

References

- [1] Anthony C. Robinson and. 2011. Highlighting in Geovisualization. *Cartography and Geographic Information Science* 38, 4 (2011), 373–383. doi:10/gd88gj
- [2] Jean-Michel Boucheix and Richard K. Lowe. 2010. An eye tracking comparison of external pointing cues and internal continuous cues in learning with complex animations. *Learning and Instruction* 20, 2 (2010), 123–135. doi:10/b935sv
- [3] Nikolaos Chatziantoniou, Akimi Oyanagi, Kenichiro Ito, Kazuma Aoyama, Hideaki Kuzuoka, and Tomohiro Amemiya. 2023. Inducing joint attention between users by visual guidance with blur effects. In *2023 IEEE Conference on Virtual Reality and 3D User Interfaces Abstracts and Workshops (VRW)*. IEEE, 789–790. doi:10/p23t
- [4] Yangming Cheng, Liulei Li, Yuanyou Xu, Xiaodi Li, Zongxin Yang, Wenguan Wang, and Yi Yang. 2023. Segment and Track Anything. *arXiv preprint arXiv:2305.06558* (2023).
- [5] Fabien Danieau, Antoine Guillo, and Renaud Doré. 2017. Attention guidance for immersive video content in head-mounted displays. In *2017 IEEE Virtual Reality (VR)*. 205–206. doi:10/p23x
- [6] Björn B. de Koning, Huib K. Tabbers, Remy M.J.P. Rikers, and Fred Paas. 2010. Attention guidance in learning from a complex animation: Seeing is understanding? *Learning and Instruction* 20, 2 (2010), 111–122. doi:10/br624h
- [7] Nina Doerr, Katrin Angerbauer, Melissa Reinelt, and Michael Sedlmair. 2023. Bees, Birds and Butterflies: Investigating the Influence of Distractors on Visual Attention Guidance Techniques. In *Extended Abstracts of the 2023 CHI Conference on Human Factors in Computing Systems* (Hamburg, Germany) (CHI EA '23). Association for Computing Machinery, New York, NY, USA, Article 51, 7 pages. doi:10/gt38bn
- [8] James T. Enns and Sarah C. MacDonald. 2013. The role of clarity and blur in guiding visual attention in photographs. *Journal of Experimental Psychology: Human Perception and Performance* 39, 2 (2013), 568–578. doi:10/f4r8xr
- [9] Nicholas Gaspelin and Steven J. Luck. 2018. The Role of Inhibition in Avoiding Distraction by Salient Stimuli. *Trends in Cognitive Sciences* 22, 1 (2018), 79–92. doi:10/gcr98d
- [10] Sandra G. Hart. 2006. Nasa-Task Load Index (NASA-TLX); 20 Years Later. *Proceedings of the Human Factors and Ergonomics Society Annual Meeting* 50, 9 (2006), 904–908. doi:10/fzvt4d
- [11] Frouke Hermens and Sunčica Zdravković. 2015. Information extraction from shadowed regions in images: An eye movement study. *Vision Research* 113 (2015), 87–96. doi:10/f7nndj
- [12] Jari K. Hietanen, Jukka M. Leppänen, Lauri Nummenmaa, and Piia Astikainen. 2008. Visuospatial attention shifts by gaze and arrow cues: An ERP study. *Brain Research* 1215 (2008), 123–136. doi:10/fbgp95
- [13] Yue Hu, Fang Qi, Yanling Wu, and Pengpeng Yang and. 2024. How textual and picture cues influence motor skill learning in short instructional videos: Evidence from eye-tracking experiments. *Journal of Psychology in Africa* 34, 4 (2024), 395–404. doi:10/p24c
- [14] Laurent Itti. 2000. *Models of bottom-up and top-down visual attention*. Ph.D. Dissertation. USA. Advisor(s) Koch, Christof.
- [15] Laurent Itti and Christof Koch. 2001. Computational modelling of visual attention. *Nature Reviews Neuroscience* 2, 3 (2001), 194–203. doi:10/chw2bk
- [16] Halszka Jarodzka, Tamara van Gog, Michael Dorr, Katharina Scheiter, and Peter Gerjets. 2013. Learning to see: Guiding students' attention via a Model's eye movements fosters learning. *Learning and Instruction* 25 (2013), 62–70. doi:10/gfs2vj
- [17] William A. Johnston and Veronica J. Dark. 1986. Selective attention. *Annual Review of Psychology* 37 (1986), 43–75. doi:10/fwhmbf
- [18] Alexander Kirillov, Eric Mintun, Nikhila Ravi, Hanzi Mao, Chloe Rolland, Laura Gustafson, Tete Xiao, Spencer Whitehead, Alexander C. Berg, Wan-Yen Lo, Piotr Dollár, and Ross Girshick. 2023. Segment Anything. *arXiv:2304.02643* (2023).
- [19] Mika Koivisto, Jukka Hyönä, and Antti Revonsuo. 2004. The effects of eye movements, spatial attention, and stimulus features on inattention blindness. *Vision Research* 44, 27 (2004), 3211–3221. doi:10/c42r3r
- [20] Janet E. Frick Krisztina Varga Jakobsen and Elizabeth A. Simpson. 2013. Look Here! The Development of Attentional Orienting to Symbolic Cues. *Journal of Cognition and Development* 14, 2 (2013), 229–249. doi:10/p23v
- [21] K. Kurzhals, M. Höferlin, and D. Weiskopf. 2013. Evaluation of Attention-Guiding Video Visualization. *Computer Graphics Forum* 32, 3pt1 (2013), 51–60. doi:10/g35jbr
- [22] Gabriel A. de Lara, Philipp N. Knechtges, Walter Paulus, and Andrea Antal. 2017. Anodal tDCS Over the Left DLPFC Did Not Affect the Encoding and Retrieval of Verbal Declarative Information. *Frontiers in Neuroscience* Volume 11 - 2017 (2017). doi:10/gf939b
- [23] Jieun Lee, Nahyun Lee, Jangkyu Ju, Jihwan Chae, Jiyeon Park, Hoe Sung Ryu, and Yang Seok Cho. 2024. Visual Complexity of Head-Up Display in Automobiles Modulates Attentional Tunneling. *Human Factors* 66, 7 (2024), 1879–1892. doi:10/p237
- [24] Jiazhou Liu, Kadek Ananta Satriadi, Barrett Ens, and Tim Dwyer. 2024. Investigating the Effects of Physical Landmarks on Spatial Memory for Information Visualisation in Augmented Reality. In *2024 IEEE International Symposium on Mixed and Augmented Reality (ISMAR)*. 289–298. doi:10/p232
- [25] Arien Mack. 2003. Inattention Blindness: Looking Without Seeing. *Current Directions in Psychological Science* 12, 5 (2003), 180–184. doi:10/bfvbhx
- [26] Victor A. Mateescu and Ivan V. Bajić. 2014. Attention Retargeting by Color Manipulation in Images. In *Proceedings of the 1st International Workshop on Perception Inspired Video Processing* (Orlando, Florida, USA) (PIVP '14). Association for Computing Machinery, New York, NY, USA, 15–20. doi:10/p235
- [27] NASCAR. 2022. 2020 Daytona 500 | NASCAR Full Race Replay. YouTube. <https://www.youtube.com/watch?v=OblznptjKDc> Video.
- [28] Peng Peng, Kai-Fu Yang, Si-Qin Liang, and Yong-Jie Li. 2022. Contour-guided saliency detection with long-range interactions. *Neurocomputing* 488 (2022), 345–358. doi:10/p23z
- [29] Emma Jane Pretty, Renan Luigi Martins Guarese, Haytham M. Fayek, and Fabio Zambetta. 2024. Comparing Subjective Measures of Workload in Video Game Play: Evaluating the Test-Retest Reliability of the VGDS and NASA-TLX. In *Companion Proceedings of the 26th International Conference on Multimodal Interaction* (San Jose, Costa Rica) (ICMI Companion '24). Association for Computing Machinery, New York, NY, USA, 56–60. doi:10/g8vrbq
- [30] Nathan D. Rasmussen, Daniel R. Thornton, and Bryan S. Morse. 2008. Enhancement of unusual color in aerial video sequences for assisting wilderness search and rescue. In *2008 15th IEEE International Conference on Image Processing*. 1356–1359. doi:10/ckn8js
- [31] Anastasia Schmitz, Andrew MacQuarrie, Simon Julier, Nicola Binetti, and Anthony Steed. 2020. Directing versus Attracting Attention: Exploring the Effectiveness of Central and Peripheral Cues in Panoramic Videos. In *2020 IEEE Conference on Virtual Reality and 3D User Interfaces (VR)*. 63–72. doi:10/gktm6j
- [32] Alessandra Semeraro and Laia Turmo Vidal. 2022. Visualizing Instructions for Physical Training: Exploring Visual Cues to Support Movement Learning from Instructional Videos. In *Proceedings of the 2022 CHI Conference on Human Factors in Computing Systems* (New Orleans, LA, USA) (CHI '22). Association for Computing Machinery, New York, NY, USA, Article 90, 16 pages. doi:10/gp3g8m
- [33] Daniel J Simons and Christopher F Chabris. 1999. Gorillas in Our Midst: Sustained Inattention Blindness for Dynamic Events. *Perception* 28, 9 (1999), 1059–1074. doi:10/gdh8td
- [34] Norman J. Slamecka and Brian McEree. 1983. Normal forgetting of verbal lists as a function of their degree of learning. *Journal of Experimental Psychology: Learning, Memory, and Cognition* 9, 3 (1983), 384–397. doi:10/dpjm5
- [35] Jonathan Sutton, Tobias Langlotz, Alexander Plopski, and Kasper Hornbæk. 2024. Flicker Augmentations: Rapid Brightness Modulation for Real-World Visual Guidance using Augmented Reality. In *Proceedings of the 2024 CHI Conference on Human Factors in Computing Systems* (Honolulu, HI, USA) (CHI '24). Association for Computing Machinery, New York, NY, USA, Article 752, 19 pages. doi:10/p23r
- [36] Jonathan Sutton, Tobias Langlotz, Alexander Plopski, Stefanie Zollmann, Yuta Itoh, and Holger Regenbrecht. 2022. Look over there! Investigating Saliency Modulation for Visual Guidance with Augmented Reality Glasses. In *Proceedings of the 35th Annual ACM Symposium on User Interface Software and Technology* (Bend, OR, USA) (UIST '22). Association for Computing Machinery, New York, NY, USA, Article 81, 15 pages. doi:10/gq7jgb
- [37] Massimo Turatto and Giovanni Galfano. 2000. Color, form and luminance capture attention in visual search. *Vision Research* 40, 13 (2000), 1639–1643. doi:10/fh3q2m
- [38] Manuela Waldner, Mathieu Le Muzic, Matthias Bernhard, Werner Purgathofer, and Ivan Viola. 2014. Attractive Flicker — Guiding Attention in Dynamic Narrative Visualizations. *IEEE Transactions on Visualization and Computer Graphics* 20, 12 (2014), 2456–2465. doi:10/f6qhz4
- [39] Xue Wang, Lin Lin, Meiqi Han, and J. Michael Spector. 2020. Impacts of cues on learning: Using eye-tracking technologies to examine the functions and designs of added cues in short instructional videos. *Computers in Human Behavior* 107 (2020), 106279. doi:10/gnw32q
- [40] Christopher D. Wickens and Amy L. Alexander and. 2009. Attentional Tunneling and Task Management in Synthetic Vision Displays. *The International Journal of Aviation Psychology* 19, 2 (2009), 182–199. doi:10/cf9vsd
- [41] Zongxin Yang and Yi Yang. 2022. Decoupling Features in Hierarchical Propagation for Video Object Segmentation. In *Advances in Neural Information Processing Systems (NeurIPS)*.